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# Understanding and applying coulombic efficiency in lithium metal batteries

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**Coulombic efficiency (CE) has been widely used in battery research as a quantifiable indicator for the reversibility of batteries. While CE helps to predict the lifespan of a lithium-ion battery, the prediction is not necessarily accurate in a rechargeable lithium metal battery. Here, we discuss the fundamental definition of CE and unravel its true meaning in lithium-ion batteries and a few representative configurations of lithium metal batteries. Through examining the similarities and differences of CE in lithium-ion batteries and lithium metal batteries, we establish a CE measuring protocol with the aim of developing high-energy long-lasting practical lithium metal batteries. The understanding of CE and the CE protocol are broadly applicable in other rechargeable metal batteries including Zn, Mg and Na batteries.**

Batteries that can offer superior performance over the lithium-ion battery (LIB) counterpart, or so called beyond LIB technologies, for various applications such as electrical vehicles and grid energy storage are urgently needed. Among many beyond-LIBs, rechargeable lithium metal batteries (LMBs) with the potential to double the energy density of state-of-art LIBs have been extensively investigated. Recent results have demonstrated promising progress[1] with hundreds of stable cycles achieved in realistic high-energy pouch cells[2].

The vast majority of laboratory research has been conducted with coin cells. While coin cells are useful to offer fundamental insights of chemical and electrochemical processes, performance data generated from coin cell studies are problematic. This is because many of the key parameters in coin cells are often not even in the same magnitude as those acceptable in realistic high-energy batteries[3]. This inconsistency in testing conditions leads to many confusing performance results and greatly decelerates research innovation in meaningful directions. The same problem also arises in sodium, magnesium and zinc batteries, in which unrealitic mass/volume ratios between the anode, cathode and electrolyte in coin cells lead to the same reproducibility issues in realistic batteries.

We have previously explored the minimum requirements of major cell components in building high-energy Li metal pouch cells[1] and proposed coin cell testing conditions for LMBs. However, another important attribute of LMBs, the cycling stability, is always challenging to predict. Coulombic efficiency (CE) is usually used to estimate the cycling life of LIB because CE reflects the loss of $Li^+$ during each cycle. Although CE has also been frequently linked to explain the Li metal cycle life, it is not very effective to monitor the lifespan of a LMB. Even CE itself measured in different groups varies significantly due to the lack of a standard protocol relevant to the realistic high energy LMB conditions.

In this work, we examine CE and its relation to cycle life in both the LIB and several representative LMBs. The electrochemical principles that govern the applicability of CE as a cell cycling monitor are discussed. A protocol of measuring and applying CE in LMBs is also suggested.

## Fundementals of CE and its applicability in LIBs

By definition, CE is the ratio of discharge capacity over charge capacity of a specific electrode in a cell. Because capacity is measured by the total charge flow to/from the electrode and the total capacity of LIBs is usually cathode-limiting, CE can be expressed as the ratio between the amount of $Li^+$ ions or electrons returning to cathode and the amount of $Li^+$ or electrons departing from cathode in a full cycle, as indicated in equation (1).

$$\begin{aligned} \mathrm{CE} &= \frac{\text{Discharge capacity}}{\text{Charge capacity}} \\ &= \frac{\text{Total number of } Li^+ \text{ ions back to cathode}}{\text{Total number of } Li^+ \text{ ions departing from cathode}} \\ &= \frac{\text{Total number of } e^- \text{ back to cathode}}{\text{Total number of } e^- \text{ departing from cathode}} \end{aligned} \quad (1)$$

Assuming no side reactions on the electrodes in an ideal cell, the flow of $Li^+$ or electrons should be completely from reversible electrochemical reactions. CE then equals to 100% according to equation (1). In realistic cells, however, side reactions between electrolytes and electrodes are ubiquitous. The side reactions can be either chemical or electrochemical. In the former case, electrons generated from the parasitic reactions may or may not be collected by the current collectors, depending on the reaction pathways. In the latter case, electron loss and acceptance still take place on the current collectors but are irreversible. In particular, some electrolyte decomposition such as electrolyte polymerization could lead to

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the formation of byproducts that are not electrochemically active[4–6], and thus become irreversible. The electrons released from those irreversible electrochemical reactions can be acquired by the current collector and subsequently counted into the CE calculation, which complicates the meaning of the obtained CE.

In a conventional LIB (Fig. 1a), there is no Li metal. Graphite is dominantly used as the anode. All $Li^+$ ions are originally stored in the cathode side in LIBs. During charge, $Li^+$ ions de-intercalate from the cathode (for example, $LiNi_xMn_yCo_zO_2$; NMC, $x + y + z = 1$) lattice and insert into the layered structure of graphite (Fig. 1a). Some $Li^+$ cations are irreversibly consumed for the solid electrolyte interphase (SEI) formation on graphite surfaces during charge. Thereby, in the subsequent discharge process, fewer $Li^+$ ions return into the NMC structure compared to the amount of $Li^+$ ions that leave from the NMC. Accordingly, CE is less than 100%.

If electrolytes such as ethylene carbonate and vinylene carbonate are used in a LIB, they are known to form a stable SEI film against graphite anodes, and the irreversible consumption of $Li^+$ ions will likely become negligible after the formation cycles, that is, the first few cycles at a very slow rate in order to form stable SEI layers. Furthermore, if the Graphite/NMC cell is operated at 4.25 V or below, after the formation cycles, the electrochemical reactions become quite stable with minimum side reactions. This means that the CE also becomes stable thereafter. The majority of electrons in the CE definition (equation 1) can be attributed to the reversible electrochemical reactions and equals the number of $Li^+$ ions involved in the redox reaction, which directly determines the reversible capacity of the subsequent cycling. Therefore, the remaining capacity in the cell after certain cycling (Fig.1b) can be effectively calculated by using the conventional method (equation 2)[7–9].

$$\text{Capacity retention} = (\text{CE})^n \tag{2}$$

Where CE refers to coulombic efficiency and $n$ is the cycle number. If 1000 stable cycles with more than 90% capacity retention is desired, the averaged CE would have to be at least 99.99%.

If the charge voltage of the Graphite/NMC battery is raised to beyond 4.25 V, the cathode electrolyte interphase (CEI) and sometimes the reorganization of the cathode lattice also begin to consume $Li^+$ or alter the number of reversible $Li^+$ ions in the system. The presence of shuttle reactions will also alter CE and thus complicate the relationship between CE and cell cycle life[10]. CE measurements also necessitate high precision along with temperature and pressure control, if an accurate prediction of long-term cycling is desired[11,12]. To apply equation (2), an optimized electrolyte recipe, stable electrodes and consistent testing conditions are all necessary.

## Li metal cells

While equation (2) works reasonably well for LIBs, the capacity retention prediction from CE is questionable for LMBs. We show in Figs 1c and 1d that a higher CE does not necessarily correspond to a more stable cycle life.

In a LMB, Li is not only stored in the cathode but also available from Li metal anode (Fig. 1e). During the first charge, some of the $Li^+$ ions originated from the cathode will unavoidably participate in side reactions at the anode side, resulting in a Li loss. Upon discharge, although fewer original $Li^+$ ions return to the cathode, additional 'fresh' $Li^+$ are stripped from the Li metal 'reservoir' to compensate for the partial loss of the cathode-derived $Li^+$ ions. Therefore, CE, calculated as the ratio of the amount of $Li^+$ ions returning to the cathode versus the amount of $Li^+$ ions departing from cathode in a full cycle, cannot reflect the net Li loss during each cycle. As long as there are plenty of $Li^+$ sources on the Li anode and a sufficient amount of electrolytes, cell cycling will always be able to continue regardless of the value of CE. This explains why an average CE of only 99.69% still maintains at 94% capacity after 100 cycles (Fig. 1c), although theoretically, capacity retention should decrease to only 74% based on equation (2). The same rules apply to any Li metal half cell in which Li metal is used as the anode (reference electrode) to study electrode materials such as Si for LIBs. Therefore, CE in those half cells has no direct relation to the eventual cycling performance of LIBs consisting of the same electrode material, although the half cell provides sufficient Li source to help to assess the individual electrode material itself.

Of note, CE does not always reflect the qualities of SEI and/or CEI. For example, CE in Fig. 1d is higher than in Fig. 1c in the first 20 cycles. But the cell in Fig. 1d lasts a much shorter time than that in Fig. 1c. Although a higher CE is seen in Fig. 1d, the highly resistant SEI film (and sometimes the CEI film as well) derived from carbonate-based electrolyte (Fig. 1d) noticeably accelerates the cell degradation. How long a Li metal cell stably cycles is therefore determined by the depletion rate of Li metal and electrolyte as well as the build-up rate of cell impedance, whichever comes first[3]. The morphology and surface area of deposited Li also impact the consumption rate of electrolyte and the Li[2], although the origin of uneven Li metal deposition is from mass transport in the electrolyte[13]. It is not surprising that, in some cases, Li dendrites form at a CE of close to 100%, which suddenly short the cell internally without any predictive signs[14]. Another important scenario is that sometimes a 'soft short' happens inside the cell but will not immediately short the cell and impacts CE measured. If a 'soft short' occurs inside a Li/NMC metal cell, the charge curve can be extended to a random length depending on how long the soft short lasts. The cell may still be able to discharge, but the CE will be noticeably lower than those in other cycles. If the soft short happens in Li/Li symmetric cells, it is usually reflected by the 'reduced' over-potential for both Li stripping and plating which is in fact a stable internal electronic connection between two Li electrodes[15]. Therefore, CE does not necessarily represent reversibility of the Li metal cell which is determined by many factors including CE.

## Anode-free cells

A Li metal cell without the Li metal anode bears some resemblance to a Li-ion cell in that the Li source is only available from the cathode (Fig. 1f). The copper current collector at the anode side often features rough surfaces with inhomogeneous properties (Fig. 2a), leading to the loss of a large amount of Li during each cycle. Figure 2b suggests that even at a shallow deposition of 0.1 mAh $cm^{-2}$, Li shows preferential nucleation locations instead of homogenously covering the copper surface. In anode-free cells, the deposited Li anode is completely from cathodes. Charged cathodes are facing a thin but very uneven Li deposited on the anode. Therefore, after the first charge, the ratio of areal capacities between nagetive and positive electrode (referred as N/P ratio) in the cell varies significantly at different locations, which may cause uneven utilization of cathodes and thus disastrous capacity loss in realistic batteries. The first stripping of Li from Cu foil also results in a very uneven surface morphology (Fig. 2c), accelerating the uneven distribution of the electrical field at the anode side and worsening Li deposition in the subsequent cycles.

Figure 2d shows a typical cycling and corresponding CE of an anode-free cell. The areal capacity of NMC used is ~4 mAh $cm^{-2}$, consistent with all other tests in this work. In this configuration, the averaged CE of Cu/NMC coin cell is 97.79% in 100 cycles. By using equation (2), after 100 cycles, the capacity retention of Cu/NMC cell should be about 10.7%, close to the experimentally measured value of 13.1% after 100 cycles.

In anode-free cells, the SEI accumulation on the anode side can significantly increase the cell impedance and thus strongly affect the cycling. In this regard, this unstable SEI layer formed on the anode side is quite different from the traditional SEI in LIBs and is more preferably called reaction or passivation layers instead. For example, when a carbonate-based electrolyte is used, a very low CE

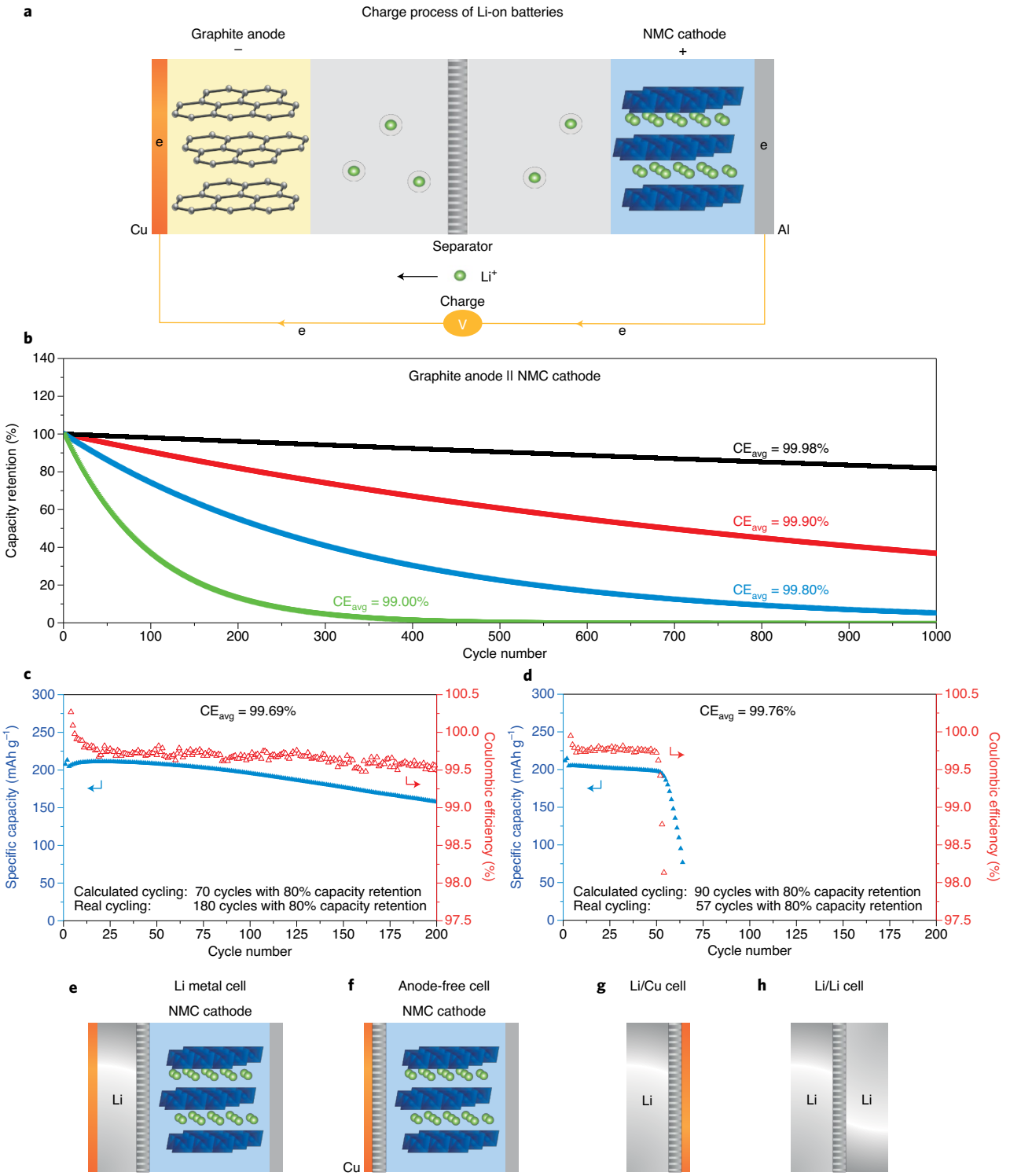


**Fig. 1 | Various cell configurations and comparisons between CEs and capacity retentions. a**, Charge process of Li-ion batteries (cut-off voltage, 4.25 V). **b**, Cycling of Li-ion batteries with different CE values. Their cycling performances are consistent with prediction from averaged CE based on equation (2). **c**,**d**, Cycling performances and CEs of Li || NMC811 coin cells in two different electrolytes. The cell in **c** has an average CE of 99.69% and cycled stably for more than 180 cycles. The cell in **d** has a much higher average CE of 99.76% but lasts only for 50 cycles. **c** is tested in compatible electrolyte comprised of 1.2 M LiFSI in TEP/BTFE (1:3 by molar ratio), while **d** is tested in conventional carbonate electrolyte comprised of 1.0 M $LiPF_6$ in EC/EMC (3:7 by weight) with 2 wt% VC. The comparison clearly indicates that CE cannot be directly used in Li metal cells to evaluate electrolytes and predict cell cycle life. Both cells after two formation cycles at C/10 were cycled at room temperature by charging at C/10 rate to 4.4 V and at 4.4 V to C/20, and then discharging at C/3 rate to 2.7 V. Li thickness, 50 μm; NMC loading, 4.0 mAh $cm^{-2}$. **e**–**h**, Different configurations of Li metal cells used in research.

of 92.31% is observed (Fig. 2e) for the limited 40 cycles. Compared to the morphology of Li deposited in a more compatible electrolyte (Fig. 2f), Li deposited in the carbonate-based electrolyte (Fig. 2g) is more porous accommodating more parasitic reactions. Therefore, the insulating passivation layers quickly build up, increase cell impedance and eventually terminate cell operation.